

able at the centre provide an opportunity to carry out both experimental research and design and modelling research for nanoelectronic devices. There are several groups in the institute, working on various aspects of synthesis of new materials, processing techniques, device structures, modelling and simulation.

Current scope in India

Talking about how this sector is poised to grow in India, Nirav Kothari, founder and owner, Nilima Nanotechnologies, informs that it is at a very nascent stage. He says, "All the activity that is happening is only at the laboratory scale, with not very large levels of funding. This is restricting innovation, thereby reducing lab-to-marketplace products. However, the innovations that are actually made do possess potential. Many gaps need to be filled in though."

On a similar note, Krishna Kant, managing director, AVANSA Technology & Services says, "At present, it is in very early stages. It is growing rapidly and will be big in coming years."

Surinder Bhagat, country HR head, Freescale Semiconductor India Pvt Ltd, adds that, "We are yet to see students coming with the specialisation in nanoelectronics, but I maintain that there are huge opportunities in the sector and is yet to boom in India."

Further adding on the scope for a career in India, Nirav says that, "In India, there isn't much unless you are employed with a blue-chip firm with a dedicated interest in nanoelectronics, where constant R&D is performed, allowing a hands-on approach for engineers desiring to work in the field. There aren't many 'nano jobs' available."

Educational requirements

Nanotechnology is mostly involved with properties and interactions of matter, and thus physics, chemistry and biology all come into play. However, nanoelectronics is an interdisciplinary field where electronics, engineering and science intersect.

Top nanotechnology colleges in India

- IITs at Mumbai, Kanpur, Chennai, Guwahati and Delhi
- Amity Institute of Nanotechnology (AINT), Noida
- Biosys Biotech Lab & Research Centre, Chennai
- Indian Institute of Science, Bangalore
- National Institute of Technology (NIT), Rourkela
- National Institute of Technology (NIT), Kurukshetra (Haryana)
- Maulana Azad National Institute of Technology (MANIT), Bhopal
- Amrita Centre for Nanosciences (ACNS), Kochi
- Jawaharlal Nehru Center for Advanced Scientific Research, Bangalore
- Centre for Nanotechnology & Advanced Biomaterials (CeNTAB), Sastra University, Tamil Nadu

Top global institutions for nanotechnology

- State University of New York (Albany), USA
- TU Dresden (Technische Universität Dresden), Germany
- Imperial College, UK (Master's degree by research)
- University College of London, UK (MSc in nanotechnology)



Centers of Excellence in Nanoelectronics (CEN) at IISc, Bengaluru

Studying science or engineering, and paying attention to the developments in electronics and semiconductors, can provide you with a solid foundation for any broad range of careers.

Career options are available for those who hold a bachelors, masters or a Ph.D. degree in the sciences. Kothari explains, "I would suppose a minimum of a Ph.D. is required. Given that the Indian market is way too nascent with almost no funding from the government, it's generally pretty hard to sustain oneself unless you are not an academician. If, of course, an individual chooses this field, they should have a strong product, usable by the masses to start up with."

Just a handful of roles available in India

Most of the jobs available in India are that of sales engineer or an academician since nanoelectronics has negligible presence in India, informs Nirav Kothari. There are just a handful of jobs as scientists or engineers

or research officers. Commenting on the job roles in India, he says, "They could be hired as lab engineers who would perform hands-on research of a potential application of a nanoparticle for use in industry."

Talking about roles available in his company, Krishna Kant shares, "Initially, they can work as a junior research trainee if they have interest in research, else as marketing executives where there is an opportunity to work in building company's relations and deal with high-end buyers of the nanomaterial products." He further adds, "At present, we encourage young graduates to come up with new ideas and development of the products; we can provide them a platform to execute their ideas. We are a start-up company in this field and require young enthusiastic people who can join us as a team and provide their inputs in marketing as well as in the research." ●

The author is a technical correspondent at EFY Bengaluru. (Excerpts from <http://www.nano.iisc.ernet.in>)